

Texas Mathematics TEKS

Grades 3, 4, and 5

Current official standards • Verified September 20, 2026

This reference reproduces the official knowledge and skills and student expectations from 19 TAC §§111.5–111.7, with grade-specific codes added for easy reference. Each grade begins on a new page. Mathematical process standards and personal financial literacy are included.

Version note

TEA currently lists these elementary mathematics standards as “Adopted 2012.” The linked elementary rules PDF is labeled “December 2014 Update.” These dates identify the current published rules; they do not mean that newer grade 3–5 math standards were omitted.

How the codes work

For example, **3.2D** means Grade 3, knowledge and skills statement (2), student expectation (D). Standards **5.5** and **5.7** have no lettered subparts in the official rules and retain those codes. Wording and punctuation are preserved; only numbering and layout are reformatted.

Official sources

[TEA: 19 TAC Chapter 111 — Mathematics](#)

[TEA: Subchapter A — Elementary Mathematics \(official PDF\)](#)

Source sections: §111.5 Grade 3; §111.6 Grade 4; §111.7 Grade 5. Each was adopted to be effective September 10, 2012, 37 TexReg 7109. This is a reformatted reference, not a TEA-issued document.

Grade 3 Mathematics

19 TAC §111.5 • Adopted 2012 • Knowledge and skills

3.1 | Mathematical process standards

The student uses mathematical processes to acquire and demonstrate mathematical understanding. The student is expected to:

- 3.1A** apply mathematics to problems arising in everyday life, society, and the workplace;
- 3.1B** use a problem-solving model that incorporates analyzing given information, formulating a plan or strategy, determining a solution, justifying the solution, and evaluating the problem-solving process and the reasonableness of the solution;
- 3.1C** select tools, including real objects, manipulatives, paper and pencil, and technology as appropriate, and techniques, including mental math, estimation, and number sense as appropriate, to solve problems;
- 3.1D** communicate mathematical ideas, reasoning, and their implications using multiple representations, including symbols, diagrams, graphs, and language as appropriate;
- 3.1E** create and use representations to organize, record, and communicate mathematical ideas;
- 3.1F** analyze mathematical relationships to connect and communicate mathematical ideas; and
- 3.1G** display, explain, and justify mathematical ideas and arguments using precise mathematical language in written or oral communication.

3.2 | Number and operations

The student applies mathematical process standards to represent and compare whole numbers and understand relationships related to place value. The student is expected to:

- 3.2A** compose and decompose numbers up to 100,000 as a sum of so many ten thousands, so many thousands, so many hundreds, so many tens, and so many ones using objects, pictorial models, and numbers, including expanded notation as appropriate;
- 3.2B** describe the mathematical relationships found in the base-10 place value system through the hundred thousands place;
- 3.2C** represent a number on a number line as being between two consecutive multiples of 10; 100; 1,000; or 10,000 and use words to describe relative size of numbers in order to round whole numbers; and
- 3.2D** compare and order whole numbers up to 100,000 and represent comparisons using the symbols $>$, $<$, or $=$.

3.3 | Number and operations

The student applies mathematical process standards to represent and explain fractional units. The student is expected to:

- 3.3A** represent fractions greater than zero and less than or equal to one with denominators of 2, 3, 4, 6, and 8 using concrete objects and pictorial models, including strip diagrams and number lines;
- 3.3B** determine the corresponding fraction greater than zero and less than or equal to one with denominators of 2, 3, 4, 6, and 8 given a specified point on a number line;
- 3.3C** explain that the unit fraction $\frac{1}{b}$ represents the quantity formed by one part of a whole that has been partitioned into b equal parts where b is a non-zero whole number;
- 3.3D** compose and decompose a fraction $\frac{a}{b}$ with a numerator greater than zero and less than or equal to b as a sum of parts $\frac{1}{b}$;
- 3.3E** solve problems involving partitioning an object or a set of objects among two or more recipients using pictorial representations of fractions with denominators of 2, 3, 4, 6, and 8;
- 3.3F** represent equivalent fractions with denominators of 2, 3, 4, 6, and 8 using a variety of objects and pictorial models, including number lines;
- 3.3G** explain that two fractions are equivalent if and only if they are both represented by the same point on the number line or represent the same portion of a same size whole for an area model; and
- 3.3H** compare two fractions having the same numerator or denominator in problems by reasoning about their sizes and justifying the conclusion using symbols, words, objects, and pictorial models.

3.4 | Number and operations

The student applies mathematical process standards to develop and use strategies and methods for whole number computations in order to solve problems with efficiency and accuracy. The student is expected to:

- 3.4A** solve with fluency one-step and two-step problems involving addition and subtraction within 1,000 using strategies based on place value, properties of operations, and the relationship between addition and subtraction;
- 3.4B** round to the nearest 10 or 100 or use compatible numbers to estimate solutions to addition and subtraction problems;
- 3.4C** determine the value of a collection of coins and bills;
- 3.4D** determine the total number of objects when equally-sized groups of objects are combined or arranged in arrays up to 10 by 10;
- 3.4E** represent multiplication facts by using a variety of approaches such as repeated addition, equal-sized groups, arrays, area models, equal jumps on a number line, and skip counting;
- 3.4F** recall facts to multiply up to 10 by 10 with automaticity and recall the corresponding division facts;
- 3.4G** use strategies and algorithms, including the standard algorithm, to multiply a two-digit number by a one-digit number. Strategies may include mental math, partial products, and the commutative, associative, and distributive properties;
- 3.4H** determine the number of objects in each group when a set of objects is partitioned into equal shares or a set of objects is shared equally;

3.4I determine if a number is even or odd using divisibility rules;

3.4J determine a quotient using the relationship between multiplication and division; and

3.4K solve one-step and two-step problems involving multiplication and division within 100 using strategies based on objects; pictorial models, including arrays, area models, and equal groups; properties of operations; or recall of facts.

3.5 | Algebraic reasoning

The student applies mathematical process standards to analyze and create patterns and relationships. The student is expected to:

3.5A represent one- and two-step problems involving addition and subtraction of whole numbers to 1,000 using pictorial models, number lines, and equations;

3.5B represent and solve one- and two-step multiplication and division problems within 100 using arrays, strip diagrams, and equations;

3.5C describe a multiplication expression as a comparison such as 3×24 represents 3 times as much as 24;

3.5D determine the unknown whole number in a multiplication or division equation relating three whole numbers when the unknown is either a missing factor or product; and

3.5E represent real-world relationships using number pairs in a table and verbal descriptions.

3.6 | Geometry and measurement

The student applies mathematical process standards to analyze attributes of two-dimensional geometric figures to develop generalizations about their properties. The student is expected to:

3.6A classify and sort two- and three-dimensional figures, including cones, cylinders, spheres, triangular and rectangular prisms, and cubes, based on attributes using formal geometric language;

3.6B use attributes to recognize rhombuses, parallelograms, trapezoids, rectangles, and squares as examples of quadrilaterals and draw examples of quadrilaterals that do not belong to any of these subcategories;

3.6C determine the area of rectangles with whole number side lengths in problems using multiplication related to the number of rows times the number of unit squares in each row;

3.6D decompose composite figures formed by rectangles into non-overlapping rectangles to determine the area of the original figure using the additive property of area; and

3.6E decompose two congruent two-dimensional figures into parts with equal areas and express the area of each part as a unit fraction of the whole and recognize that equal shares of identical wholes need not have the same shape.

3.7 | Geometry and measurement

The student applies mathematical process standards to select appropriate units, strategies, and tools to solve problems involving customary and metric measurement. The student is expected to:

3.7A represent fractions of halves, fourths, and eighths as distances from zero on a number line;

3.7B determine the perimeter of a polygon or a missing length when given perimeter and remaining side lengths in problems;

3.7C determine the solutions to problems involving addition and subtraction of time intervals in minutes using pictorial models or tools such as a 15-minute event plus a 30-minute event equals 45 minutes;

3.7D determine when it is appropriate to use measurements of liquid volume (capacity) or weight; and

3.7E determine liquid volume (capacity) or weight using appropriate units and tools.

3.8 | Data analysis

The student applies mathematical process standards to solve problems by collecting, organizing, displaying, and interpreting data. The student is expected to:

3.8A summarize a data set with multiple categories using a frequency table, dot plot, pictograph, or bar graph with scaled intervals; and

3.8B solve one- and two-step problems using categorical data represented with a frequency table, dot plot, pictograph, or bar graph with scaled intervals.

3.9 | Personal financial literacy

The student applies mathematical process standards to manage one's financial resources effectively for lifetime financial security. The student is expected to:

3.9A explain the connection between human capital/labor and income;

3.9B describe the relationship between the availability or scarcity of resources and how that impacts cost;

3.9C identify the costs and benefits of planned and unplanned spending decisions;

3.9D explain that credit is used when wants or needs exceed the ability to pay and that it is the borrower's responsibility to pay it back to the lender, usually with interest;

3.9E list reasons to save and explain the benefit of a savings plan, including for college; and

3.9F identify decisions involving income, spending, saving, credit, and charitable giving.

Grade 4 Mathematics

19 TAC §111.6 • Adopted 2012 • Knowledge and skills

4.1 | Mathematical process standards

The student uses mathematical processes to acquire and demonstrate mathematical understanding. The student is expected to:

- 4.1A** apply mathematics to problems arising in everyday life, society, and the workplace;
- 4.1B** use a problem-solving model that incorporates analyzing given information, formulating a plan or strategy, determining a solution, justifying the solution, and evaluating the problem-solving process and the reasonableness of the solution;
- 4.1C** select tools, including real objects, manipulatives, paper and pencil, and technology as appropriate, and techniques, including mental math, estimation, and number sense as appropriate, to solve problems;
- 4.1D** communicate mathematical ideas, reasoning, and their implications using multiple representations, including symbols, diagrams, graphs, and language as appropriate;
- 4.1E** create and use representations to organize, record, and communicate mathematical ideas;
- 4.1F** analyze mathematical relationships to connect and communicate mathematical ideas; and
- 4.1G** display, explain, and justify mathematical ideas and arguments using precise mathematical language in written or oral communication.

4.2 | Number and operations

The student applies mathematical process standards to represent, compare, and order whole numbers and decimals and understand relationships related to place value. The student is expected to:

- 4.2A** interpret the value of each place-value position as 10 times the position to the right and as one-tenth of the value of the place to its left;
- 4.2B** represent the value of the digit in whole numbers through 1,000,000,000 and decimals to the hundredths using expanded notation and numerals;
- 4.2C** compare and order whole numbers to 1,000,000,000 and represent comparisons using the symbols $>$, $<$, or $=$;
- 4.2D** round whole numbers to a given place value through the hundred thousands place;
- 4.2E** represent decimals, including tenths and hundredths, using concrete and visual models and money;
- 4.2F** compare and order decimals using concrete and visual models to the hundredths;
- 4.2G** relate decimals to fractions that name tenths and hundredths; and
- 4.2H** determine the corresponding decimal to the tenths or hundredths place of a specified point on a number line.

4.3 | Number and operations

The student applies mathematical process standards to represent and generate fractions to solve problems. The student is expected to:

- 4.3A** represent a fraction a/b as a sum of fractions $1/b$, where a and b are whole numbers and $b > 0$, including when $a > b$;
- 4.3B** decompose a fraction in more than one way into a sum of fractions with the same denominator using concrete and pictorial models and recording results with symbolic representations;
- 4.3C** determine if two given fractions are equivalent using a variety of methods;
- 4.3D** compare two fractions with different numerators and different denominators and represent the comparison using the symbols $>$, $=$, or $<$;
- 4.3E** represent and solve addition and subtraction of fractions with equal denominators using objects and pictorial models that build to the number line and properties of operations;
- 4.3F** evaluate the reasonableness of sums and differences of fractions using benchmark fractions 0, $1/4$, $1/2$, $3/4$, and 1, referring to the same whole; and
- 4.3G** represent fractions and decimals to the tenths or hundredths as distances from zero on a number line.

4.4 | Number and operations

The student applies mathematical process standards to develop and use strategies and methods for whole number computations and decimal sums and differences in order to solve problems with efficiency and accuracy. The student is expected to:

- 4.4A** add and subtract whole numbers and decimals to the hundredths place using the standard algorithm;
- 4.4B** determine products of a number and 10 or 100 using properties of operations and place value understandings;
- 4.4C** represent the product of 2 two-digit numbers using arrays, area models, or equations, including perfect squares through 15 by 15;
- 4.4D** use strategies and algorithms, including the standard algorithm, to multiply up to a four-digit number by a one-digit number and to multiply a two-digit number by a two-digit number. Strategies may include mental math, partial products, and the commutative, associative, and distributive properties;
- 4.4E** represent the quotient of up to a four-digit whole number divided by a one-digit whole number using arrays, area models, or equations;
- 4.4F** use strategies and algorithms, including the standard algorithm, to divide up to a four-digit dividend by a one-digit divisor;
- 4.4G** round to the nearest 10, 100, or 1,000 or use compatible numbers to estimate solutions involving whole numbers; and
- 4.4H** solve with fluency one- and two-step problems involving multiplication and division, including interpreting remainders.

4.5 | Algebraic reasoning

The student applies mathematical process standards to develop concepts of expressions and equations. The student is expected to:

- 4.5A** represent multi-step problems involving the four operations with whole numbers using strip diagrams and equations with a letter standing for the unknown quantity;
- 4.5B** represent problems using an input-output table and numerical expressions to generate a number pattern that follows a given rule representing the relationship of the values in the resulting sequence and their position in the sequence;
- 4.5C** use models to determine the formulas for the perimeter of a rectangle ($l + w + l + w$ or $2l + 2w$), including the special form for perimeter of a square ($4s$) and the area of a rectangle ($l \times w$); and
- 4.5D** solve problems related to perimeter and area of rectangles where dimensions are whole numbers.

4.6 | Geometry and measurement

The student applies mathematical process standards to analyze geometric attributes in order to develop generalizations about their properties. The student is expected to:

- 4.6A** identify points, lines, line segments, rays, angles, and perpendicular and parallel lines;
- 4.6B** identify and draw one or more lines of symmetry, if they exist, for a two-dimensional figure;
- 4.6C** apply knowledge of right angles to identify acute, right, and obtuse triangles; and
- 4.6D** classify two-dimensional figures based on the presence or absence of parallel or perpendicular lines or the presence or absence of angles of a specified size.

4.7 | Geometry and measurement

The student applies mathematical process standards to solve problems involving angles less than or equal to 180 degrees. The student is expected to:

- 4.7A** illustrate the measure of an angle as the part of a circle whose center is at the vertex of the angle that is "cut out" by the rays of the angle. Angle measures are limited to whole numbers;
- 4.7B** illustrate degrees as the units used to measure an angle, where $\frac{1}{360}$ of any circle is one degree and an angle that "cuts" $\frac{n}{360}$ out of any circle whose center is at the angle's vertex has a measure of n degrees. Angle measures are limited to whole numbers;
- 4.7C** determine the approximate measures of angles in degrees to the nearest whole number using a protractor;
- 4.7D** draw an angle with a given measure; and
- 4.7E** determine the measure of an unknown angle formed by two non-overlapping adjacent angles given one or both angle measures.

4.8 | Geometry and measurement

The student applies mathematical process standards to select appropriate customary and metric units, strategies, and tools to solve problems involving measurement. The student is expected to:

- 4.8A** identify relative sizes of measurement units within the customary and metric systems;
- 4.8B** convert measurements within the same measurement system, customary or metric, from a smaller unit into a larger unit or a larger unit into a smaller unit when given other equivalent measures represented in a table; and
- 4.8C** solve problems that deal with measurements of length, intervals of time, liquid volumes, mass, and money using addition, subtraction, multiplication, or division as appropriate.

4.9 | Data analysis

The student applies mathematical process standards to solve problems by collecting, organizing, displaying, and interpreting data. The student is expected to:

- 4.9A** represent data on a frequency table, dot plot, or stem-and-leaf plot marked with whole numbers and fractions; and
- 4.9B** solve one- and two-step problems using data in whole number, decimal, and fraction form in a frequency table, dot plot, or stem-and-leaf plot.

4.10 | Personal financial literacy

The student applies mathematical process standards to manage one's financial resources effectively for lifetime financial security. The student is expected to:

- 4.10A** distinguish between fixed and variable expenses;
- 4.10B** calculate profit in a given situation;
- 4.10C** compare the advantages and disadvantages of various savings options;
- 4.10D** describe how to allocate a weekly allowance among spending; saving, including for college; and sharing; and
- 4.10E** describe the basic purpose of financial institutions, including keeping money safe, borrowing money, and lending.

Grade 5 Mathematics

19 TAC §111.7 • Adopted 2012 • Knowledge and skills

5.1 | Mathematical process standards

The student uses mathematical processes to acquire and demonstrate mathematical understanding. The student is expected to:

- 5.1A** apply mathematics to problems arising in everyday life, society, and the workplace;
- 5.1B** use a problem-solving model that incorporates analyzing given information, formulating a plan or strategy, determining a solution, justifying the solution, and evaluating the problem-solving process and the reasonableness of the solution;
- 5.1C** select tools, including real objects, manipulatives, paper and pencil, and technology as appropriate, and techniques, including mental math, estimation, and number sense as appropriate, to solve problems;
- 5.1D** communicate mathematical ideas, reasoning, and their implications using multiple representations, including symbols, diagrams, graphs, and language as appropriate;
- 5.1E** create and use representations to organize, record, and communicate mathematical ideas;
- 5.1F** analyze mathematical relationships to connect and communicate mathematical ideas; and
- 5.1G** display, explain, and justify mathematical ideas and arguments using precise mathematical language in written or oral communication.

5.2 | Number and operations

The student applies mathematical process standards to represent, compare, and order positive rational numbers and understand relationships as related to place value. The student is expected to:

- 5.2A** represent the value of the digit in decimals through the thousandths using expanded notation and numerals;
- 5.2B** compare and order two decimals to thousandths and represent comparisons using the symbols $>$, $<$, or $=$; and
- 5.2C** round decimals to tenths or hundredths.

5.3 | Number and operations

The student applies mathematical process standards to develop and use strategies and methods for positive rational number computations in order to solve problems with efficiency and accuracy. The student is expected to:

- 5.3A** estimate to determine solutions to mathematical and real-world problems involving addition, subtraction, multiplication, or division;
- 5.3B** multiply with fluency a three-digit number by a two-digit number using the standard algorithm;

- 5.3C** solve with proficiency for quotients of up to a four-digit dividend by a two-digit divisor using strategies and the standard algorithm;
- 5.3D** represent multiplication of decimals with products to the hundredths using objects and pictorial models, including area models;
- 5.3E** solve for products of decimals to the hundredths, including situations involving money, using strategies based on place-value understandings, properties of operations, and the relationship to the multiplication of whole numbers;
- 5.3F** represent quotients of decimals to the hundredths, up to four-digit dividends and two-digit whole number divisors, using objects and pictorial models, including area models;
- 5.3G** solve for quotients of decimals to the hundredths, up to four-digit dividends and two-digit whole number divisors, using strategies and algorithms, including the standard algorithm;
- 5.3H** represent and solve addition and subtraction of fractions with unequal denominators referring to the same whole using objects and pictorial models and properties of operations;
- 5.3I** represent and solve multiplication of a whole number and a fraction that refers to the same whole using objects and pictorial models, including area models;
- 5.3J** represent division of a unit fraction by a whole number and the division of a whole number by a unit fraction such as $\frac{1}{3} \div 7$ and $7 \div \frac{1}{3}$ using objects and pictorial models, including area models;
- 5.3K** add and subtract positive rational numbers fluently; and
- 5.3L** divide whole numbers by unit fractions and unit fractions by whole numbers.

5.4 | Algebraic reasoning

The student applies mathematical process standards to develop concepts of expressions and equations. The student is expected to:

- 5.4A** identify prime and composite numbers;
- 5.4B** represent and solve multi-step problems involving the four operations with whole numbers using equations with a letter standing for the unknown quantity;
- 5.4C** generate a numerical pattern when given a rule in the form $y = ax$ or $y = x + a$ and graph;
- 5.4D** recognize the difference between additive and multiplicative numerical patterns given in a table or graph;
- 5.4E** describe the meaning of parentheses and brackets in a numeric expression;
- 5.4F** simplify numerical expressions that do not involve exponents, including up to two levels of grouping;
- 5.4G** use concrete objects and pictorial models to develop the formulas for the volume of a rectangular prism, including the special form for a cube ($V = l \times w \times h$, $V = s \times s \times s$, and $V = Bh$); and
- 5.4H** represent and solve problems related to perimeter and/or area and related to volume.

5.5 | Geometry and measurement

The student applies mathematical process standards to classify two-dimensional figures by attributes and properties. The student is expected to classify two-dimensional figures in a hierarchy of sets and subsets using graphic organizers based on their attributes and properties.

5.6 | Geometry and measurement

The student applies mathematical process standards to understand, recognize, and quantify volume. The student is expected to:

5.6A recognize a cube with side length of one unit as a unit cube having one cubic unit of volume and the volume of a three-dimensional figure as the number of unit cubes (n cubic units) needed to fill it with no gaps or overlaps if possible; and

5.6B determine the volume of a rectangular prism with whole number side lengths in problems related to the number of layers times the number of unit cubes in the area of the base.

5.7 | Geometry and measurement

The student applies mathematical process standards to select appropriate units, strategies, and tools to solve problems involving measurement. The student is expected to solve problems by calculating conversions within a measurement system, customary or metric.

5.8 | Geometry and measurement

The student applies mathematical process standards to identify locations on a coordinate plane. The student is expected to:

5.8A describe the key attributes of the coordinate plane, including perpendicular number lines (axes) where the intersection (origin) of the two lines coincides with zero on each number line and the given point $(0, 0)$; the x -coordinate, the first number in an ordered pair, indicates movement parallel to the x -axis starting at the origin; and the y -coordinate, the second number, indicates movement parallel to the y -axis starting at the origin;

5.8B describe the process for graphing ordered pairs of numbers in the first quadrant of the coordinate plane; and

5.8C graph in the first quadrant of the coordinate plane ordered pairs of numbers arising from mathematical and real-world problems, including those generated by number patterns or found in an input-output table.

5.9 | Data analysis

The student applies mathematical process standards to solve problems by collecting, organizing, displaying, and interpreting data. The student is expected to:

5.9A represent categorical data with bar graphs or frequency tables and numerical data, including data sets of measurements in fractions or decimals, with dot plots or stem-and-leaf plots;

5.9B represent discrete paired data on a scatterplot; and

5.9C solve one- and two-step problems using data from a frequency table, dot plot, bar graph, stem-and-leaf plot, or scatterplot.

5.10 | Personal financial literacy

The student applies mathematical process standards to manage one's financial resources effectively for lifetime financial security. The student is expected to:

5.10A define income tax, payroll tax, sales tax, and property tax;

5.10B explain the difference between gross income and net income;

5.10C identify the advantages and disadvantages of different methods of payment, including check, credit card, debit card, and electronic payments;

5.10D develop a system for keeping and using financial records;

5.10E describe actions that might be taken to balance a budget when expenses exceed income; and

5.10F balance a simple budget.